

Modeling Geographic Disparities in COVID-19 Vaccine Uptake

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Abstract

The COVID-19 pandemic led to one of the largest vaccination campaigns in recorded history, yet uptake differed dramatically across countries, as we saw those with high income nations vaccinated at nearly eight times the rate of low-income ones. This study examines the structural determinants of these geographic disparities using the Our World in Data COVID-19 dataset covering 163 countries across the full pandemic period. Three analytical stages were employed, namely the exploratory data analysis to understand patterns, ordinary least squares regression to identify predictors of vaccination coverage, and k-means clustering to uncover natural country groupings. The results point toward a single conclusion, which was that vaccination disparities are primarily driven by development factors such as economic capacity, human development, and demographic composition. GDP per capita and the Human Development Index (HDI) both correlate strongly with vaccination outcomes, while government stringency shows no significant association. Surprisingly, a clustering algorithm provided with no geographic information nonetheless recovered continental geography almost perfectly, with the high vaccination cluster being predominantly European and the low vaccination cluster predominantly African. These findings suggest that achieving vaccination equity in future pandemics requires long term investment in health system capacity in lower income countries rather than just relying on distribution.

Introduction

The COVID-19 pandemic sent the world into chaos as it was one of the most severe public health crises in modern history. This pandemic claimed over seven million lives while also producing more than 775 million confirmed cases worldwide by mid-2024 (Santangelo et al., 2024). In late 2020, the development of the COVID-19 vaccines began, and these quickly became the most consequential tool available to counteract this virus by reducing severe illness and death at a population scale. The world then saw the most rapid and extensive vaccination campaign ever undertaken in history. However, despite all of this, the rollout did not translate into an equal effect across the board. It became apparent from the earliest phases of the campaign that access to and uptake of vaccines differed massively across countries and regions, and this difference persisted long after supply constraints had eased in wealthier nations. Even 3 whole years after development began, by December 2023, 33% of the world was yet to be fully vaccinated, and the vaccination rate in high-income countries was almost 7.8 times higher than in low-income ones (Zhang et al., 2024). It became apparent that lower-income nations introduced vaccines later, were less likely to achieve key coverage milestones, and were slower to do so even where those milestones were eventually reached (Pesce et al., 2025). These massive disparities amongst regions meant that the virus became harder to fully stop and instead enabled the emergence of new variants and raised fundamental questions about whether the global health community is structurally equipped at all to respond to future pandemics effectively.

These questions meant that it became important to understand why vaccination uptake diverged so dramatically. Early explanations focused on a scarcity of vaccine doses and procurement inequality, and these were genuine constraints. They were the main reason why the COVAX facility was established so that these issues related to distribution could be solved at

reduced or even zero cost. However, even after supply improved, the gaps did not close, and it became increasingly clear that distribution problems did not tell the whole story. Yoo et al. (2022) found that while COVAX did direct a disproportionate share of doses toward lower-income countries, it nonetheless fell short of dealing with broader vaccine inequality. This is where it is important to understand the structural conditions that had in a way almost predetermined these inequalities in uptake. We can see this as Duroseau et al. (2023) argued that distribution mechanisms alone cannot compensate for these structural conditions that determine whether a country can effectively receive, store, administer, and achieve population uptake of vaccines. The fact that disparities persisted even after supply constraints eased is therefore a signal that something deeper was at work, and that it was something rooted in the pre-existing structural conditions that countries brought into the pandemic rather than during the rollout itself.

When we look into these structural conditions after examining the evidence, we can see that economic capacity is the dominant structural predictor. Moradpour et al. (2023) found that national income was a significant predictor of country-level vaccination coverage, with higher-income countries achieving much greater uptake across multiple time points. This was further reinforced by Nyachoti et al. (2023) as they found a strong positive association between gross national income per capita and vaccination coverage across 192 WHO member states. So as one might expect, the economic power of a country, and its overall vaccine uptake percentage went hand in hand. Moreover, Varbanova et al. (2024) confirmed that this relationship held even within the relatively similar group of European and OECD countries, where economic and healthcare resource variables remained among the most important determinants of vaccination effects. The fact that this finding remains consistent across different geographic scopes, analytical methods, and even phases of the pandemic is difficult to overlook as it suggests that

wealth does not just predict vaccine access in a narrow logistical sense. Instead, it predicts the institutional capacity, healthcare infrastructure, cold chain logistics, and administrative coordination that together determine whether a vaccination campaign can be effective or not.

Additionally, Human Development, which includes education, health system capacity, and life expectancy, adds another layer of depth to understanding these differences. Literature like Pattanshetty et al. (2023) found that factors such as healthcare worker density and political stability all contributed independently to vaccination coverage disparities, which meant that the economic effect operates through a variety of different conditions rather than through income alone. This is consistent with what Santangelo et al. (2024) describe as the interplay of socioeconomic factors, healthcare infrastructure, and institutional trust in shaping vaccine equity, and with the finding of Vardavas et al. (2023) that income, poverty, education, and social vulnerability were all consistently associated with vaccination rates across multiple national contexts. Taken together, these studies paint a picture of vaccination outcomes as the product of a broader development scale rather than any single measurable factor, which has important implications for how the problem should be modelled analytically.

These disparities, however, don't just stop here, as the demographic composition also adds another layer to the picture. We see that countries with older median populations tend to have both stronger demand for COVID-19 vaccination, given the direct relationship between age and severe disease risk, and also more engagement with formal healthcare systems through which vaccination was primarily delivered (Moradpour et al., 2023). The age factor is particularly relevant because older populations tend to seek out vaccination at higher rates regardless of how wealthy their country is, which means age carries its own explanatory weight beyond what income alone can account for. Within a country like the United States, Groom et al.

(2024) showed that age, race, income, and insurance status all produced significant differences in vaccination rates even where supply was abundant, further highlighting that structural disparities in vaccination usually reflect deeper patterns of inequality operating at multiple scales at the same time.

The role of government policy is more controversial though. Government stringency has been proposed as a predictor of vaccination outcomes on the assumption that stricter governments had more influence on their populations. However, Fox et al. (2023) found that among unvaccinated people in low and middle-income countries, approximately half showed a genuine willingness to be vaccinated but faced structural barriers rather than plain hesitancy. This implies that the relationship between policy stringency and vaccination coverage is not straightforward and may be highly context dependent. It may differ from country to country as some tend to have a stronger institutional capacity than others.

Machine learning approaches have begun to complement traditional regression methods in analyzing these patterns. One example in the literature is Nicholson et al. (2022) who showed that unsupervised clustering could recover meaningful geographic and structural groupings from COVID-19 outcome data, and Rasouli et al. (2026) extended this to a global framework showing that early rollout speed and healthcare capacity were associated with better country performance. So, from an analytical standpoint, combining both regression and clustering gives a better and clearer picture than just going for one of them as you are able capture not just the average associations but also distinct country classifications.

Although a lot of literature does exist on this because of the sheer magnitude of this pandemic, most of these studies are either geographically narrow or are limited to the early

rollout period when supply constraints were more common. Not many analyses have conducted a full cross-national comparison spanning the complete pandemic period while also looking at multiple socioeconomic and demographic predictors and combining regression modelling with unsupervised machine learning clustering.

This study addresses that gap directly. Using the Our World in Data global COVID-19 dataset covering 163 countries across the full pandemic period, this paper asks: to what extent do economic capacity, human development, demographic composition, and government policy explain geographic disparities in COVID-19 vaccine uptake, and do countries cluster into structurally distinct groups that correspond to observed continental patterns of vaccination inequality? The central argument is that these disparities are not the product of any single factor but of a development scale that advantaged wealthier and more developed nations throughout the vaccination campaign, and that identifying this on a global scale is essential for designing more equitable responses to future public health emergencies.

Method

Dataset

This study uses the Our World in Data (OWID) COVID-19 dataset which is one of the most comprehensive publicly available sources of pandemic related data in the world. This dataset aggregates official national reports from the World Health Organization, national health ministries, the United Nations Population Division, and the World Bank, and it covers approximately 200 countries across the full duration of the pandemic. The primary outcome variable is the percentage of each country's population that completed a primary vaccination series. The explanatory variables span four categories: economic capacity represented by GDP

per capita, overall development status captured by the Human Development Index, demographic structure represented by median age and the percentage of the population aged 65 and older, and government policy response measured by the Oxford COVID-19 Government Response Stringency Index summarised as each country's mean score across the full pandemic period.

Preprocessing

Because the dataset is structured as a daily panel, the most recent non-missing observation per country for each variable were taken, as a cross-sectional snapshot of the dataset. Additionally, aggregate groupings identified by ISO codes beginning with the prefix OWID were removed, and only the country level observations were retained. Also, countries with no valid observation for the primary outcome variable were excluded, leaving a final sample of 163 countries. GDP per capita was log transformed to address its strong right skew which exists because of certain countries GDP's being much higher than others. Finally, missing values were handled thoroughly, with most such values concentrated heavily among lower-income and smaller nations who had weaker statistical reporting infrastructure, ranging from 27 countries for GDP to 37 for stringency.

Analytical approach

The analysis was done in 3 main stages. First came the exploratory data analysis, to get some descriptive statistics by continent, scatter plots of vaccination against each predictor, a continental time series which was smoothed to remove reporting noise, and a Pearson correlation matrix to characterise pairwise relationships and identify potential multicollinearity among predictors.

The second stage was ordinary least squares regression where 3 models of increasing complexity were estimated. Model 1 used log GDP alone as a baseline, Model 2 adding HDI,

median age, proportion aged 65 and older, and mean stringency, and Model 3 adding continent fixed effects with Africa as the reference category. HC3 robust standard errors were used throughout, and Variance Inflation Factors (VIF) were computed to properly assess multicollinearity. Model fit was assessed using R-squared, adjusted R-squared, and the Akaike Information Criterion (AIC).

The third stage was k-means clustering using six standardised input variables: fully vaccinated percentage, log GDP, HDI, median age, proportion aged 65 and older, and mean stringency. Continent was deliberately excluded as an input. The optimal number of clusters was selected via the elbow method ($k=4$), and the solution was validated using Principal Component Analysis (PCA).

Tools and technologies

All analysis was conducted in RStudio using Quarto and the packages of note included *tidyverse* and *ggplot2* for data manipulation and visualisation, *zoo* for rolling average smoothing, *lmtest* and *sandwich* for robust standard errors, *car* for Variance Inflation Factor computation, and *factoextra* and *cluster* for clustering and PCA visualisation.

Evaluation metrics

Regression performance was evaluated using R-squared, adjusted R-squared, and AIC, with individual predictor significance assessed at the 0.05 threshold using HC3 robust standard errors. For the clustering stage, the primary validation metric was the proportion of total variance captured by the first two principal components in the PCA, with a higher proportion indicating that the two-dimensional visualisation faithfully represents the original clustering space.

Results

Geographic Variation and Trends

After the initial EDA examining the 163 countries, stark disparities became apparent as Africa recorded a mean full vaccination rate of just 35.4 percent, more than 23 percentage points below North America at 58.1 percent, which was the next lowest continent. Oceania led at 71.1 percent, followed by Europe at 66.8 percent, South America at 65.9 percent, and Asia at 65.8 percent. As shown in Figure 1, these continental differences are immediately visible, with Africa sitting entirely apart from every other region.

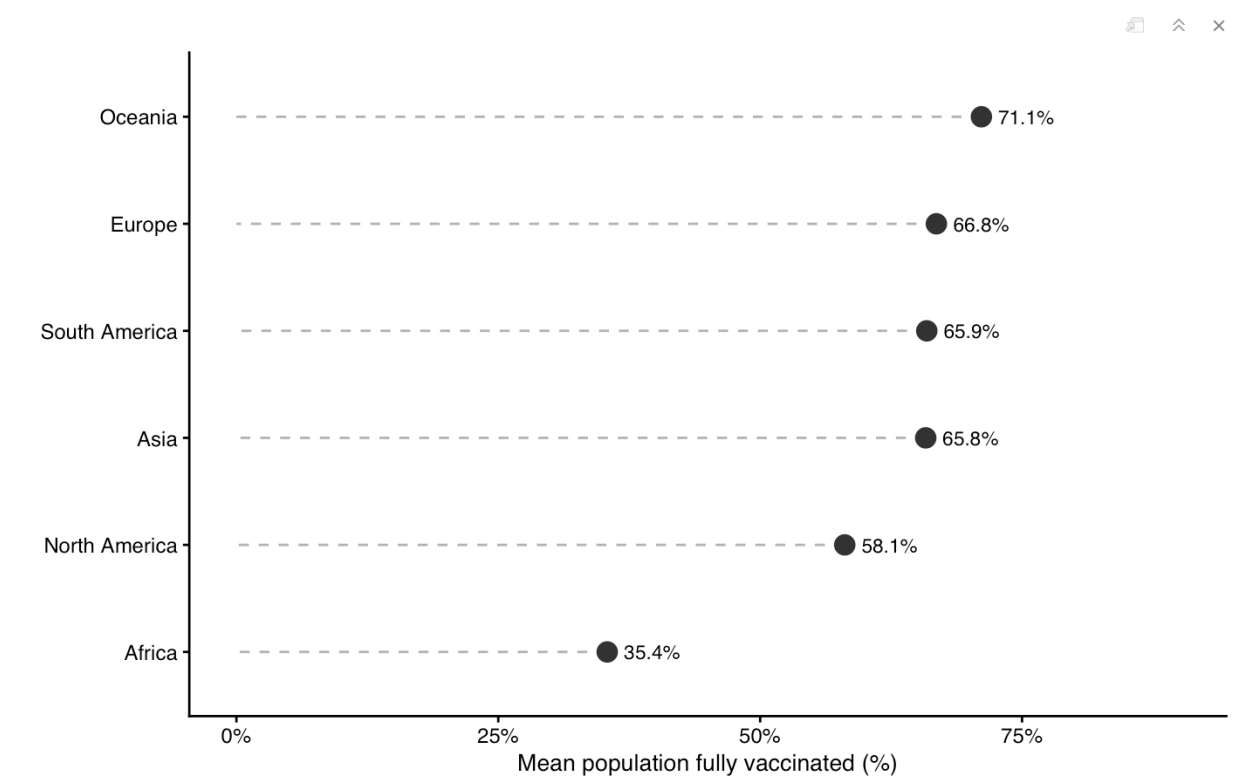


Figure-1 – *Continental dot plot*

Within-continent variation was also substantial, as we can see in Figure 2. It is apparent that Africa's distribution was concentrated at low coverage levels, while Asia and Oceania showed extremely wide spreads. This showed enormous diversity within those continental groupings.

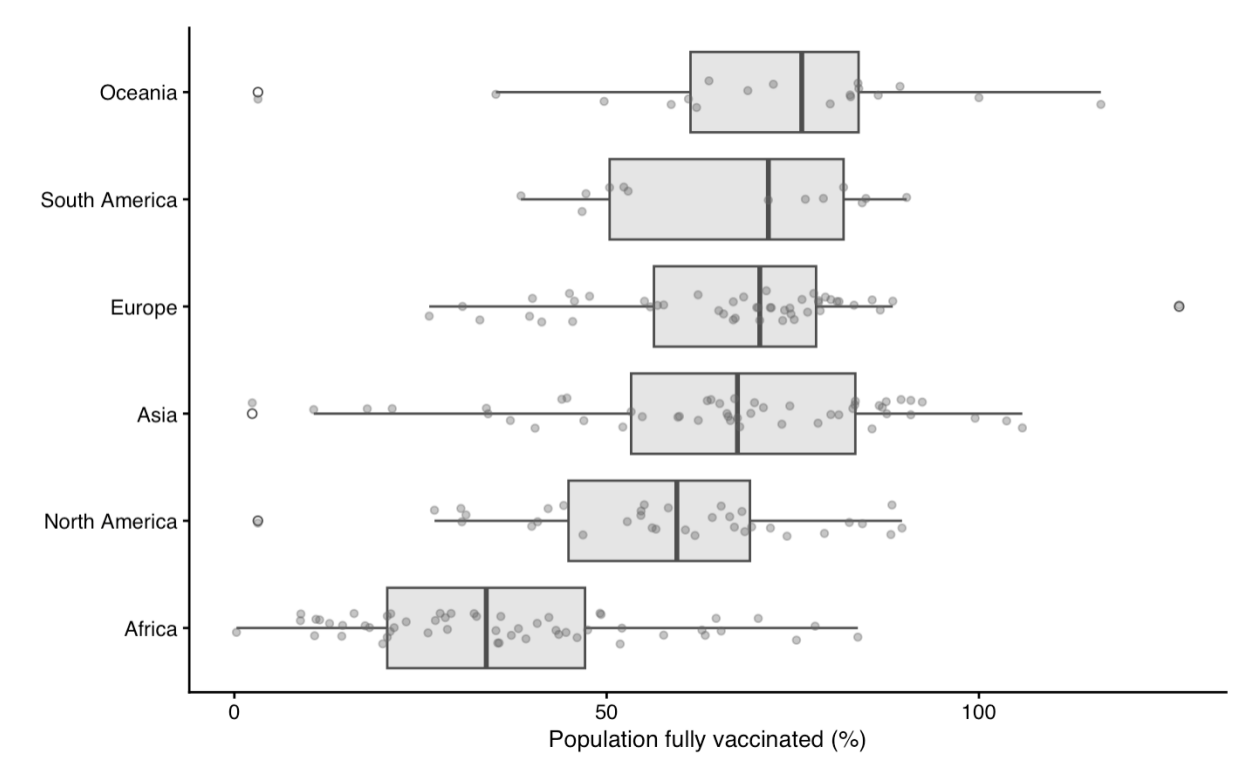


Figure 2 – *Continental box plot*

Next was the time series, presented in Figure 3, which revealed that all continents began near zero in early 2021 and diverged rapidly. Europe, Asia, South America, and North America followed steep upward trajectories, reaching plateaus between 60 and 80 percent by mid-2022, whereas Africa, on the other hand, rose at a substantially lesser scale and never approached the slope of any other continent. The gap was established within the first six months of the global

rollout and remained basically constant even afterwards, which indicated that the disparity was structural rather than a matter of timing.

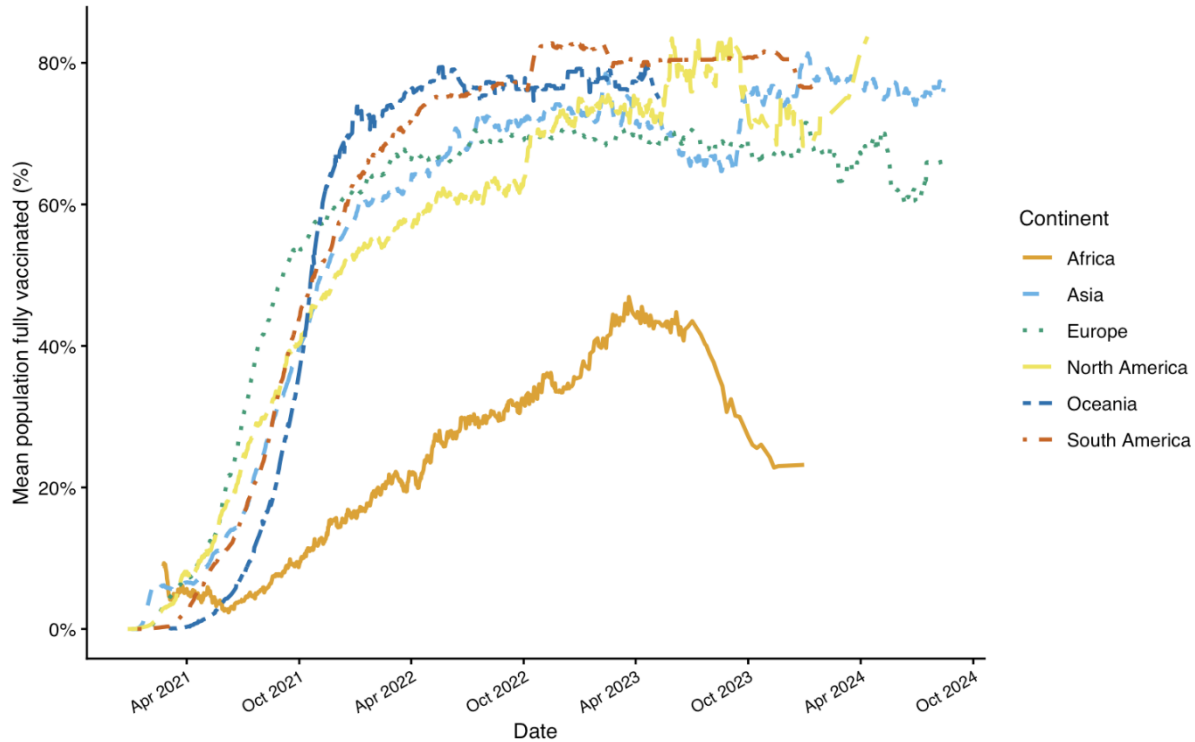


Figure 3 – *Time series*

Bivariate Relationships

The correlation matrix, as seen in Figure 4, showed that full vaccination percentage was most strongly associated with log GDP per capita at $r = 0.63$ and HDI at $r = 0.64$, with median age moderately strong at $r = 0.58$ and stringency weak at $r = 0.21$. More importantly, the predictor variables themselves were highly intercorrelated, with log GDP and HDI correlating at $r = 0.94$ and HDI and median age at $r = 0.90$. This was also the first sign of the multicollinearity findings that would be found later on in the regression analysis.

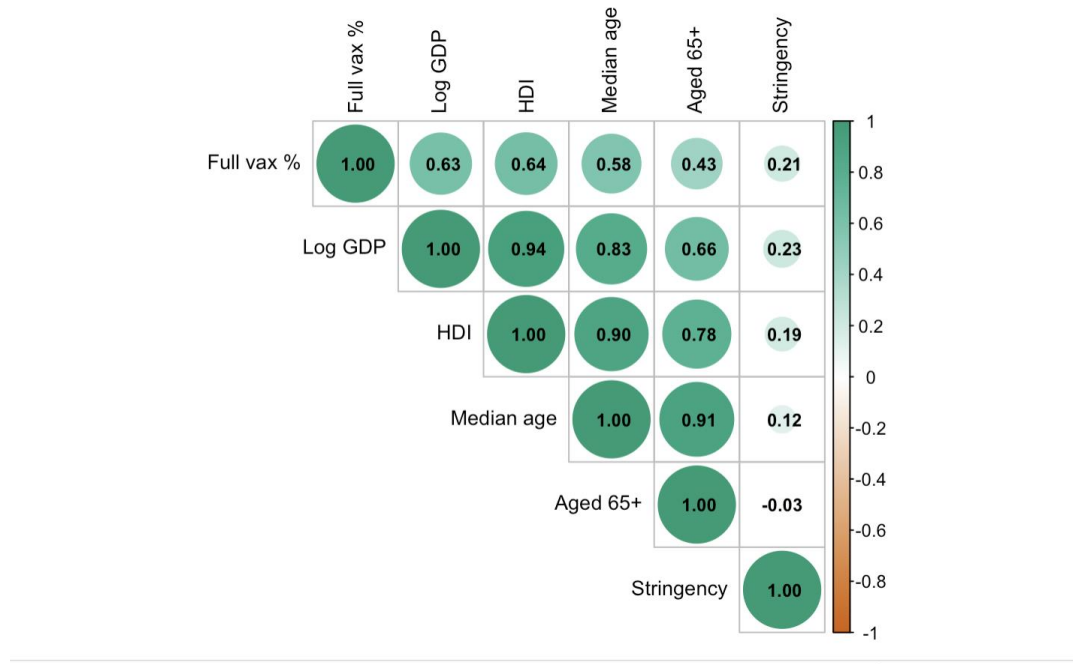


Figure 4: *Correlation matrix*

The GDP scatter plot show in in Figure 5, showing log GDP per capita on the x axis against full vaccination percentage on the y axes, confirmed the positive trend across the full income range, with African countries clustering at the bottom left and European countries at the upper right.

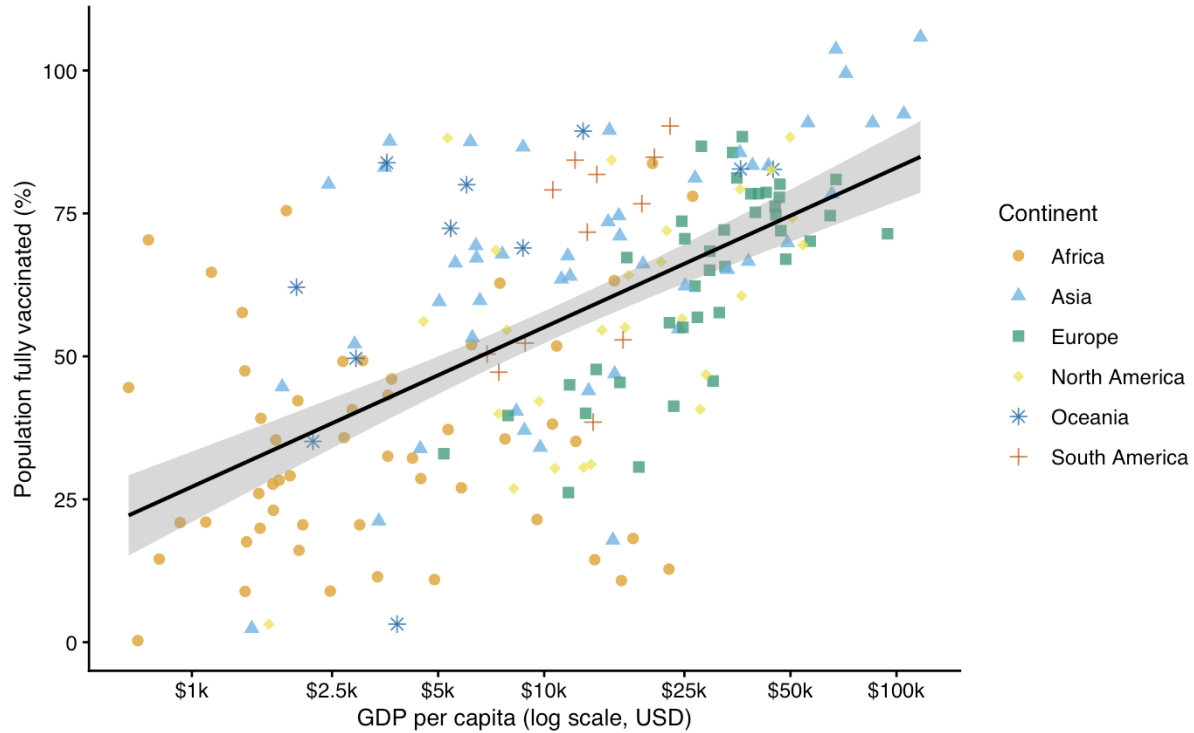


Figure 5 – *GDP Scatter plot*

Regression Results

Model 1 explained 36.3 percent of cross-national variance with an AIC of 1653.4. Model 2 increased this to 44.3 percent with an AIC of 1440.0. Model 3 achieved the best fit with R-squared of 0.497, adjusted R-squared of 0.464, and AIC of 1433.3, confirming it as the preferred specification. This can be seen in table one, which shows the model comparison table with their respective R-squared, adjusted R-squared, and AIC values side by side.

Table 1 — *Model comparison table*

Model	R2	Adj. R2	AIC
M1: GDP only	0.3627	0.3593	1653.419
M2: Full	0.4430	0.4255	1440.036
M3: Full + Continent FE	0.4967	0.4640	1433.313

In Model 3, log GDP and HDI both lost individual statistical significance, with p-values of 0.393 and 0.454 respectively under robust standard errors. VIF diagnostics confirmed severe multicollinearity, with HDI at 16.3, median age at 14.5, and log GDP at 11.1, all exceeding the threshold of 10. Median age was the only continuous predictor retaining marginal significance, with a coefficient of 1.22 and a p-value of 0.050, suggesting that each additional year of median national age is associated with approximately 1.2 percentage points higher vaccination coverage. Stringency showed no significant association in any specification, with a p-value of 0.628. Among the continent fixed effects, Asia recorded a significant positive coefficient of 12.51 relative to Africa with a p-value of 0.01. The full regression table results are shown in Table 2.

Table 2 — *Full regression results*

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-30.9880	15.8758	-1.9519	0.0528	-62.3504	0.3744
log_gdp	3.4204	3.9952	0.8561	0.3932	-4.4719	11.3128
hdi	30.1215	40.1086	0.7510	0.4538	-49.1126	109.3557
median_age	1.2196	0.6170	1.9767	0.0499	0.0007	2.4384
aged_65_older	-0.4547	0.7437	-0.6115	0.5418	-1.9238	1.0143
stringency_mean	-0.0792	0.1631	-0.4854	0.6281	-0.4015	0.2431
continentAsia	12.5136	4.8028	2.6055	0.0101	3.0257	22.0015
continentEurope	-5.8566	7.2880	-0.8036	0.4229	-20.2538	8.5407
continentNorth America	2.7984	5.9407	0.4710	0.6383	-8.9375	14.5342
continentOceania	10.0274	7.4625	1.3437	0.1810	-4.7146	24.7694

Clustering and Comparative Results

The elbow plot showed that within-cluster variance flattened more or less from $k = 4$ onward, which meant that four clusters would be selected. PCA confirmed that the first two components explained 84.1 percent of total variance, with PC1 alone accounting for 66.5 percent, indicating that the clusters are genuinely distinct rather than arbitrary. The PCA biplot in Figure 6 shows the four clusters separating cleanly along PC1 with minimal overlap.

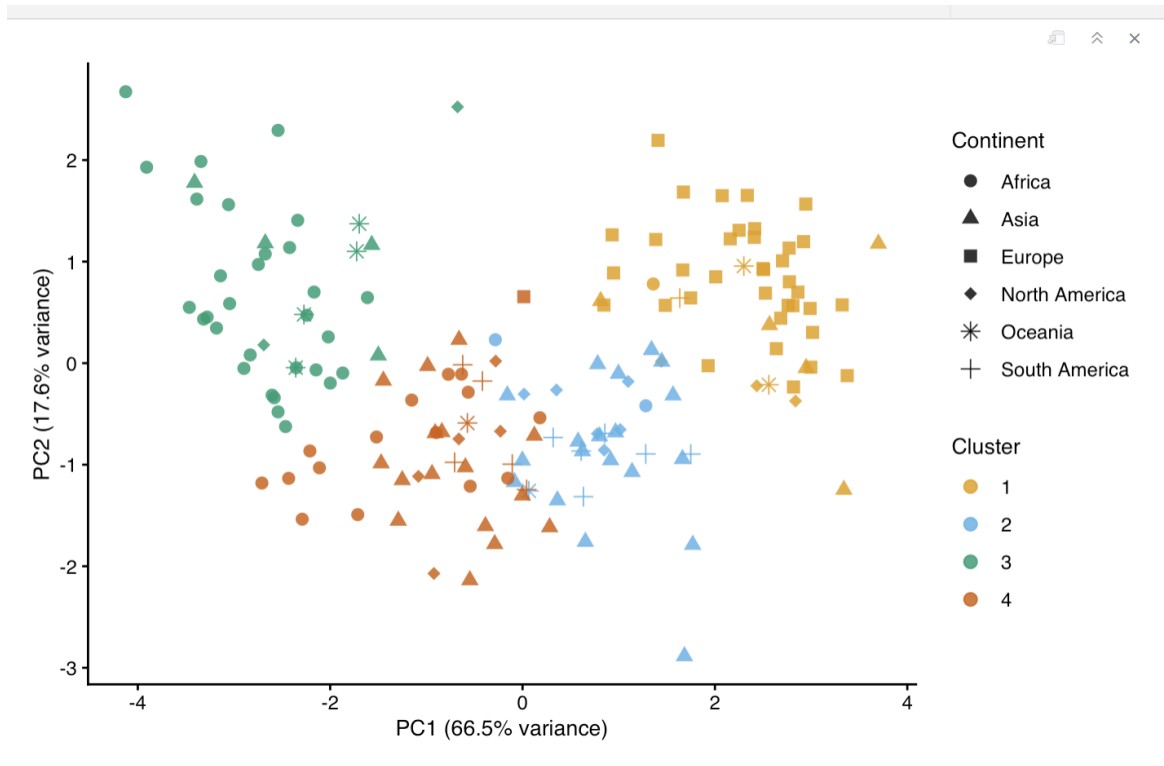
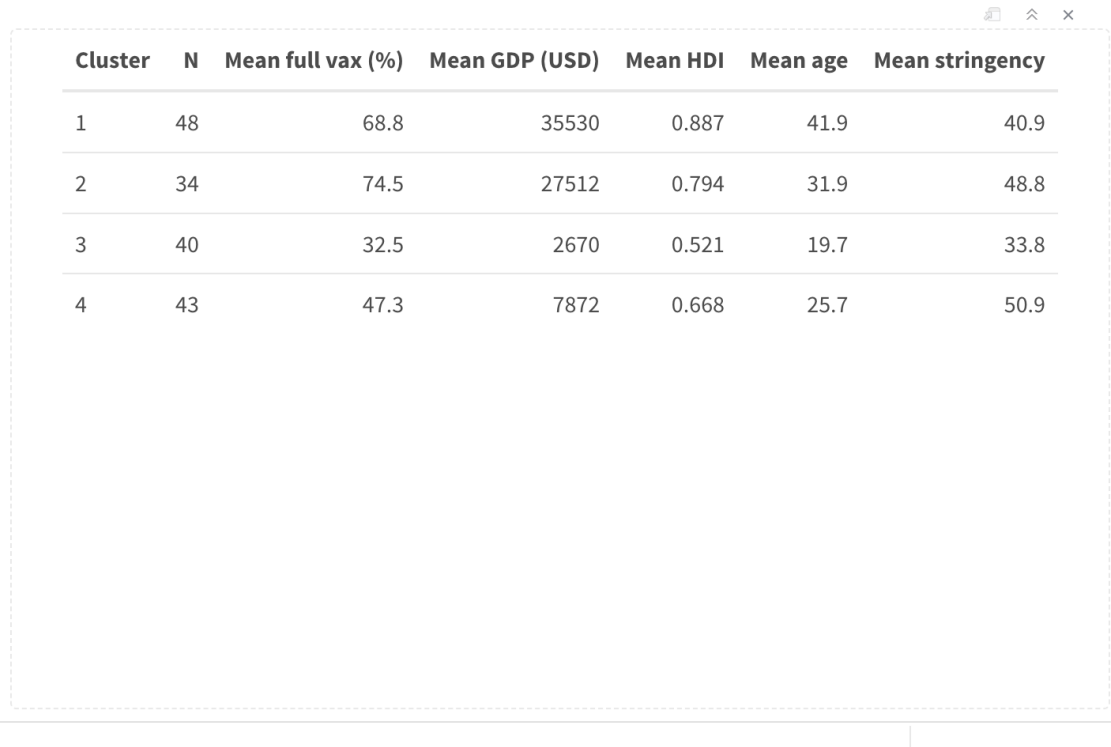


Figure 6 – PCA biplot

The four clusters differed substantially in structural profiles. Cluster 1 had an average of 68.8 percent vaccination with a mean GDP of 35,530 dollars and HDI of 0.887. Additionally, Cluster 2 achieved the highest vaccination of all at 74.5 percent even though it had lower GDP than Cluster 1 at 27,512 dollars, but it differed because of its highest mean stringency score of 48.8. Cluster 3 averaged just 32.5 percent vaccination with a mean GDP of 2,670 dollars and HDI of 0.521. Finally, Cluster 4 fell into a moderate middle position across all dimensions. We can see all these cluster profiles in Table 3.

Table 3 — *Cluster profiles*


Cluster	N	Mean full vax (%)	Mean GDP (USD)	Mean HDI	Mean age	Mean stringency
1	48	68.8	35530	0.887	41.9	40.9
2	34	74.5	27512	0.794	31.9	48.8
3	40	32.5	2670	0.521	19.7	33.8
4	43	47.3	7872	0.668	25.7	50.9

Despite continent being excluded as a clustering input, Cluster 1 was approximately 75 percent European and Cluster 3 was approximately 65 to 70 percent African in composition. Figure 7 shows this with a cluster composition chart, which shows the percentage of each continent being represented using each of the 4 clusters. This also shows how Cluster 1 is dominated by European nations, and Cluster 3 by African ones, despite their geography not being an input variable.

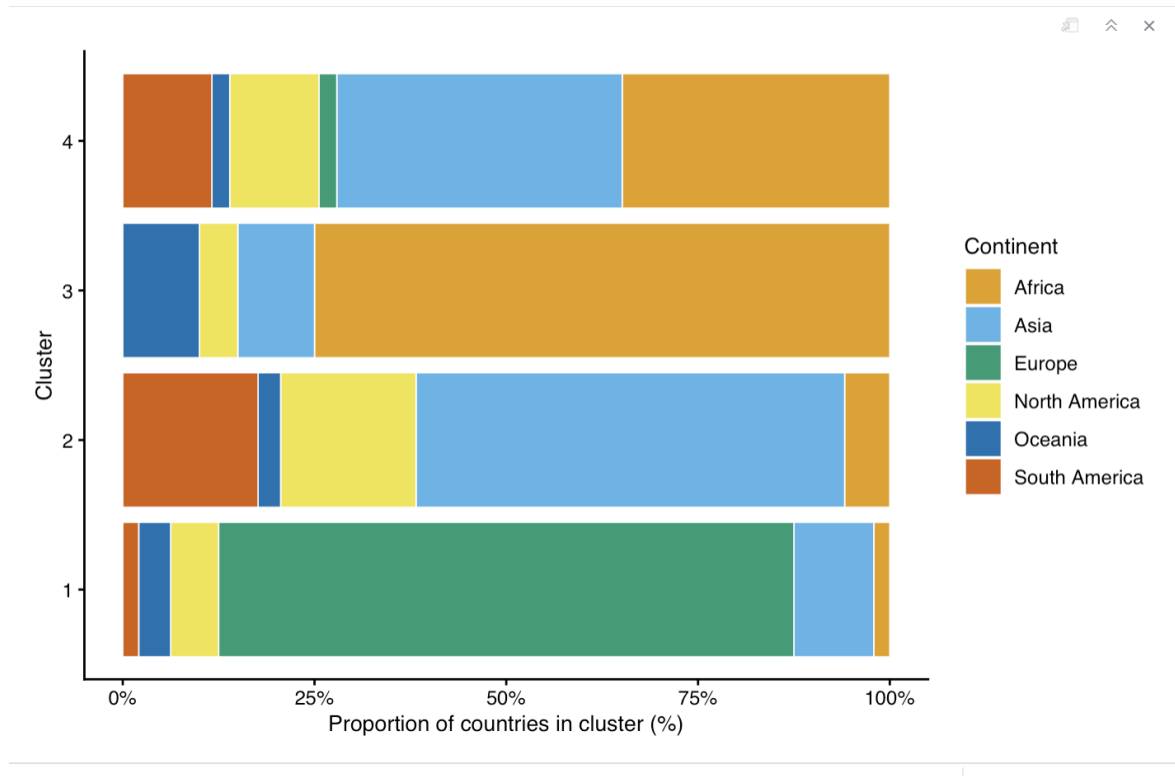


Figure 7 – Cluster Composition Chart

The top 20 and bottom 20 performing countries showed dramatic structural differences. The bottom 20 averaged 12.2 percent vaccination and a mean GDP of 6,173 dollars compared to 95.0 percent and 45,530 dollars for the top 20, representing vaccination and GDP gaps of approximately 7.8 and 7.4 times respectively. HDI averaged 0.561 for the bottom 20 and 0.813 for the top 20, and median age averaged 20.8 versus 37.2 years. These differences are shown in Table 4, which gives a comparison table showing the differences between the top vs bottom 20 countries in their GDP per capita, HDI, and median age.

Table 4 — *Top vs bottom 20 summary*

Group	Mean full vax (%)	Mean GDP (USD)	Mean HDI	Mean median age
Bottom 20	12.2	6173	0.561	20.8
Top 20	95.0	45530	0.813	37.2

Discussion

When we look at the results of this study, a consistent story keeps being brought up. In fact, even across all three distinct analytical approaches, covering exploratory data analysis, regression modelling, and unsupervised machine learning clustering, the findings keep pointing to a single conclusion. That is that the geographic disparities in COVID-19 vaccine uptake are fundamentally structural in origin. They are not the product of hesitancy, bad luck, or poor timing. They are the product of a development gradient that existed long before the first vaccine dose was administered, and that systematically determined which countries could manage an effective vaccination campaign and which could not.

The descriptive findings establish this picture clearly. Africa's mean vaccination rate of 35.4 percent stood more than 23 percentage points below the next lowest continent, and the time

series showed that this gap was not something that gradually emerged as the rollout progressed. It was there from the beginning, visible within the first six months, and never narrowed. That early and persistent divergence is one of the most important findings in this study because it tells you something specific about the nature of the problem. Gaps that form early and stay fixed are not logistical failures that better coordination could have solved. Instead, they are structural in nature, and they reflect already existing inequalities in the capacity to deliver healthcare at a population scale. This interpretation is consistent with Pesce et al. (2025), who found that lower-income countries introduced vaccines later and were systematically less likely to achieve key coverage milestones, and with Duroseau et al. (2023), who documented that the structural barriers facing low and middle-income countries during the rollout were because of long-standing deficits in healthcare infrastructure, as well as things like cold chain capacity, and institutional coordination that no distribution program operating on a pandemic timeline could realistically overcome.

The regression findings add some quantitative insight to this structural argument. We found that GDP per capita alone explains 36 percent of cross-national variance in vaccination rates, which again closely relates with Moradpour et al. (2023) and Nyachoti et al. (2023), both of whom documented strong and consistent positive associations between national income and vaccination coverage across large multi-country samples. The full model with continent fixed effects pushes this to 50 percent (R squared of 0.50), meaning that the structural variables measured here all together account for half of why countries differ in their vaccination outcomes. That is a meaningful result. It is not a complete explanation, and this study does not claim to be one, but it establishes that economic capacity, human development, and demographic composition are not peripheral factors in this story. They are central to it.

Another thing to be wary of is the multicollinearity finding, which deserves careful interpretation rather than being dismissed as an inconvenience in the analysis. The fact that GDP, HDI, and median age correlate with each other at above $r = 0.90$ is not a flaw in the data or the analysis, but instead a genuine feature of how the world is organized. We know that wealthy countries tend to have higher human development and older populations simultaneously because these conditions have complemented each other due to decades of sustained investment in sectors like healthcare and education, as well as economic growth. When regression cannot separate their individual contributions, it is because in the real world they do not operate separately. They move together as a package, and the appropriate analytical response is to interpret them collectively rather than in isolation. What the regression tells us, taken as a whole, is that the combination of these development conditions explains half of global vaccination inequality, and that trying to attribute the effect to any single variable misrepresents the nature of the problem.

Median age emerging as the only individually significant predictor in the full model is an interesting nuance worth exploring. As noted by Moradpour et al. (2023) and Viswanath et al. (2021), older populations tend to have stronger perceived personal risk from COVID-19, greater engagement with formal healthcare systems, and stronger baseline trust in medical institutions. These demand-side factors likely operate independently of pure economic capacity to some degree, meaning that demographic structure carries its own weight that is not fully absorbed by income or HDI. The age effect does not simply disappear once you account for a country's wealth, suggesting that how old a population is matters for vaccination outcomes in its own right. This finding has practical implications for how future vaccination campaigns should be designed

and prioritized, especially in settings where demographic structure and economic capacity do not move together in the way they do across much of Europe.

The government stringency finding is perhaps the most counterintuitive result in the study and the one most likely to challenge conventional assumptions. Across all three model specifications, stringency showed no statistically significant association with vaccination coverage. One might assume that stricter governments were able to control their populations more effectively, but the evidence here does not support that assumption, especially when you look at it from the global level. Again, tying it into the literature we see that it is consistent with Fox et al. (2023), who found that among unvaccinated respondents in low and middle-income countries, approximately half retained genuine willingness to be vaccinated but faced structural access barriers rather than hesitancy or resistance to policy mandates. This goes to show that if people want to be vaccinated but cannot access a facility within reasonable distance, have no cold storage infrastructure nearby, or live in a country where healthcare workers are too scarce to administer doses at a large enough scale, then even the strictest government policy does nothing to close that gap. The real bottleneck stems from structural conditions and that is what this study shows.

The most striking finding, however, came from the clustering analysis, and it is important to explore it further. For this stage of the analysis, a k-means algorithm was given six structural variables and asked to find natural groupings amongst the 163 countries. It was given no information about where any country is located. This meant that the algorithm did not have any continent labels, no geographic coordinates, no regional classifications to go off of. Yet when the resulting clusters were examined for their geographic composition, Cluster 1 was approximately 75 percent European and Cluster 3 was approximately 65 to 70 percent African. The algorithm

sorted countries by their development profiles and in doing so accidentally sorted them by continent. This is not a coincidence, and it is not a methodological artifact. It is a direct empirical demonstration that the development gradient and the geographic gradient are essentially the same thing. Continental disparities in vaccination are not geographic coincidences. They are structural regularities, and the clustering result makes that argument more forceful than any regression coefficient could.

Cluster 2 also complicates this picture, but in a productive way. This group achieved the highest mean vaccination rate of all four clusters at 74.5 percent despite having lower average GDP than Cluster 1, a younger median age, and an HDI score placing its members in the upper-middle rather than very high development band. What set Cluster 2 apart from the rest was its mean stringency score of 48.8, which was the the highest of all four groups, combined with a strong representation of Asian states with centralized governance structures. The cluster revealed that certain countries like Vietnam and Cambodia had vaccinated very high proportions of their populations even though their economic resources were well below the European average. What this implies is that governance capacity and institutional effectiveness could somewhat make up for the lack of wealth in driving vaccination outcomes, at least in some contexts. It is an important qualification to a purely economic account of the disparities, and it points toward political economy and health system organization as dimensions of the problem that deserve greater attention in future research. Abu El Kheir-Mataria et al. (2024) similarly identified political stability and health system strength as independent contributors to vaccine distribution outcomes alongside economic power, which is consistent with what Cluster 2 reveals here.

Next is the top versus bottom 20 comparison that provides the most concrete illustration of everything the analysis has established. A 7.8 times vaccination gap between the best and

worst performing countries is mirrored almost exactly by a 7.4 times GDP gap between those same two groups. So, the scale of economic inequality almost perfectly predicts the scale of vaccination inequality. The bottom 20 are almost exclusively African nations and conflict-affected states, while the top 20 include a mix of small wealthy territories, centralized Asian states, and European nations. This pattern is consistent with Ali et al. (2022) and Cheng et al. (2024), both of whom documented that the countries furthest from vaccination coverage targets were systematically those with the lowest human development and weakest health system capacity.

This study has several limitations that should be acknowledged openly. Since we took a cross-sectional snapshot of the dataset for analysis, it means that while strong associations between structural conditions and vaccination outcomes are clearly established, the causal direction cannot be fully confirmed. This makes sense because while wealthier countries vaccinated more, higher vaccination also supported faster economic recovery. This means that causality likely runs in both directions simultaneously. Panel data methods tracking changes within countries over time would be required to separate these pathways more cleanly. Additionally, the multicollinearity among development predictors means that individual coefficient estimates should be interpreted cautiously, and it makes more sense to look at all the predictors collectively. Moreover, the missing data pattern, concentrated among lower-income nations, means the true magnitude of global vaccination disparities is likely somewhat larger than what this analysis captures. And the absence of a direct measure of vaccine hesitancy means that demand-side attitudinal factors, which Moradpour et al. (2023) identified as significant particularly in higher-income settings, are not fully accounted for in the regression models.

When we look at the implications of this study, some uncomfortable realities come to light. First, the vaccination campaign made it very clear that just distributing doses is not the same as achieving vaccination equity. Although COVAX was a necessary initiative, and even Yoo et al. (2022) showed that it did succeed in directing doses toward lower-income countries at a meaningful scale, it was still operating against structural foundations that no distribution program on a pandemic timeline could overcome. The resulting problems are not ones that can be solved just by making doses available. What is actually required is sustained long-term investment in health system strengthening that begins well before any future pandemic emergency materializes. Pattanshetty et al. (2023) demonstrated that political stability, healthcare worker density, and government health expenditure all contributed independently to vaccination coverage, which means that each of those dimensions represents a potential lever for improving future pandemic preparedness in lower-income settings.

Looking ahead, several directions for future research emerge naturally from the limitations and findings of this study. Firstly, longitudinal panel data analysis which uses time-varying measures of both structural conditions and vaccination outcomes would allow stronger causal identification of the mechanisms at work, so that there is more than just an association. Moreover, direct measures of hesitancy, healthcare worker density, and cold chain infrastructure would address key omitted variables, which might be nice to have for future analyses. A subnational analysis examining age-stratified vaccination data within countries would allow the individual-level age effect to be identified more precisely than country-level median age permits. That could be a better estimate of how the two correlate. And qualitative research into the specific governance mechanisms that enabled Cluster 2 countries to outperform their structural

conditions could help convert the quantitative pattern identified here into effective policy guidance.

So, what this study ultimately shows is that the COVID-19 pandemic did not create global health inequity, it just revealed it at such a scale that it was impossible to ignore. The countries that fell behind were not dealing with a logistics problem that arose during the pandemic. Instead, they were already behind, by decades, in the structural conditions that had already sealed their fate. Closing that gap before the next pandemic arrives should be treated as a matter of global health security for all countries involved.

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